

⇒ Design Parking Lot.

→ Overview

→ Requirements

→ Class Diagram

→ Schema Design

→ Code.

⇒ Overview.
 ↗ Know
 ↘ Don't know

⇒ Command line application.

⇒ In memory DB.

Requirement Gathering.

→ Multiple parking floors.

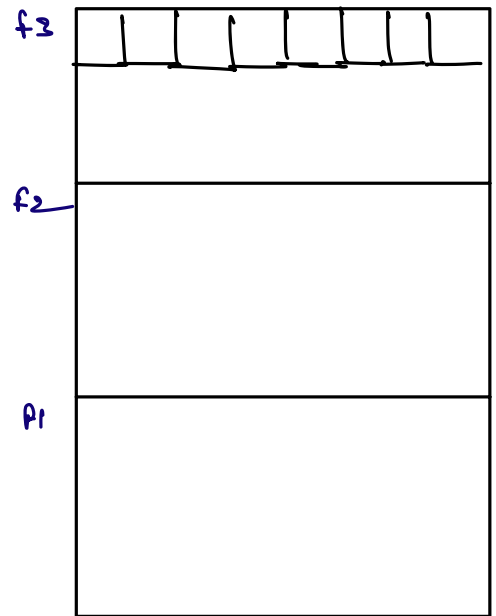
→ Diff. parking slots for diff type of vehicles.

→ A token is generated at the time of entry.

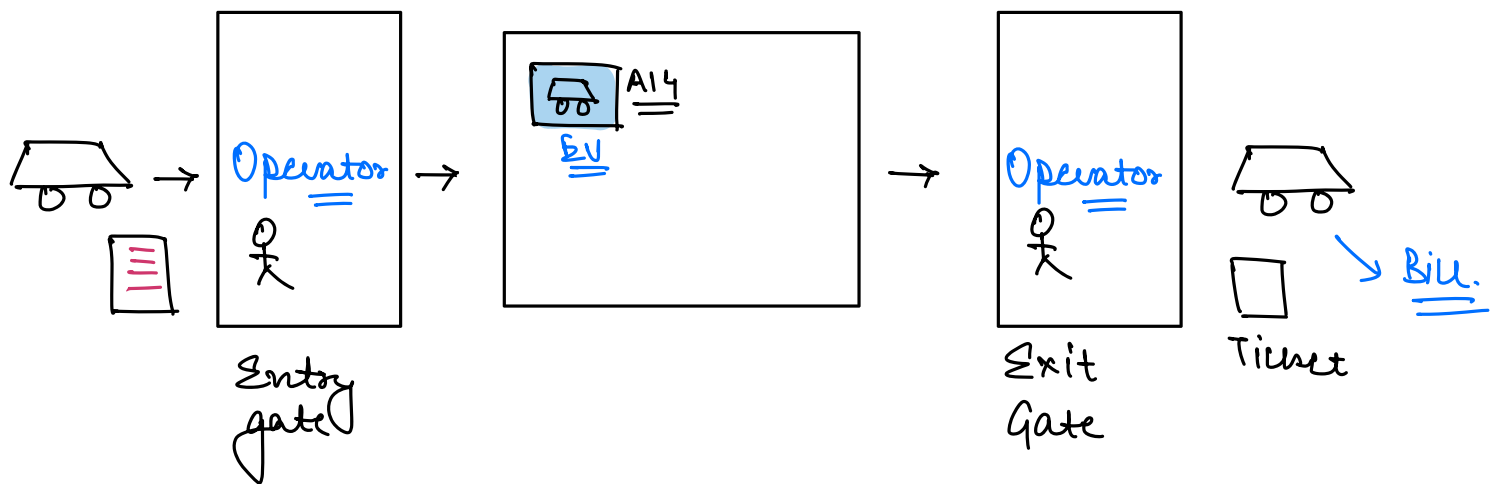
→ Bill will be generated at the time of exit.

→ Payment should be made at exit gate.

→ Parking spot will be assigned at the time of entry.



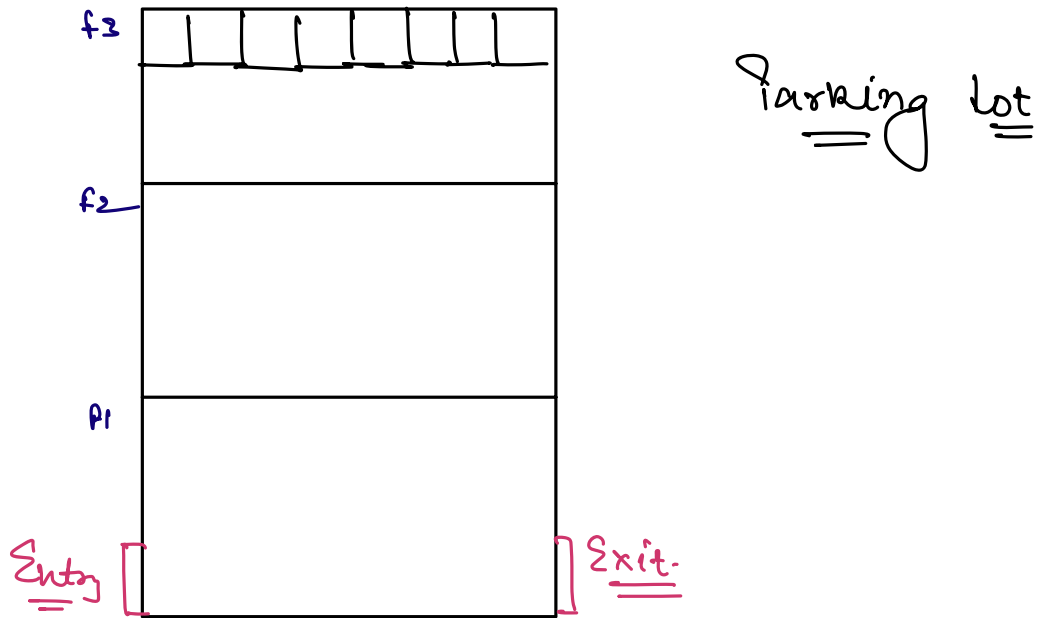
- Payment can be done online or in cash.
- Payment calculation algo can be dynamic.
- For online payments, we'll use 3rd party payment gateways.

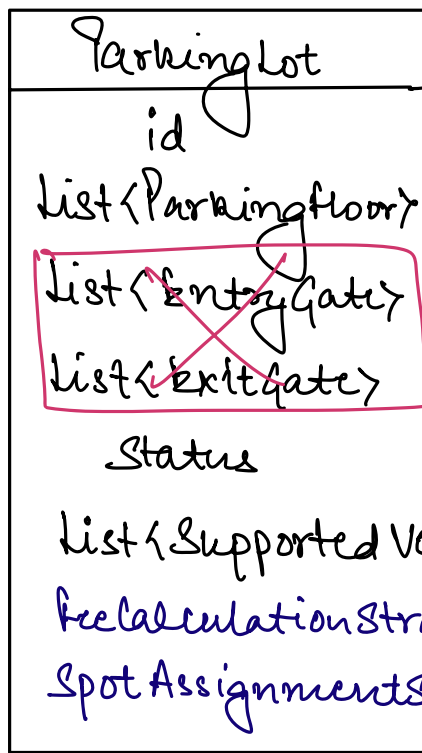


- Multiple Entry & Exit gates.
- An operator will be there at each gate.
- Different ways of assigning a parking spot.

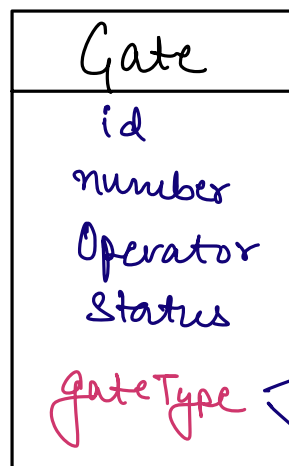
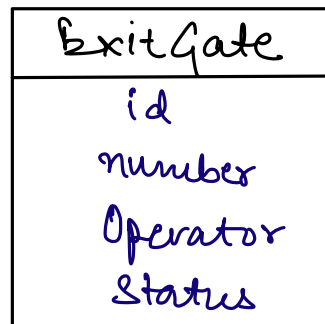
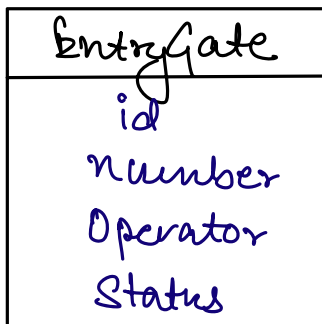
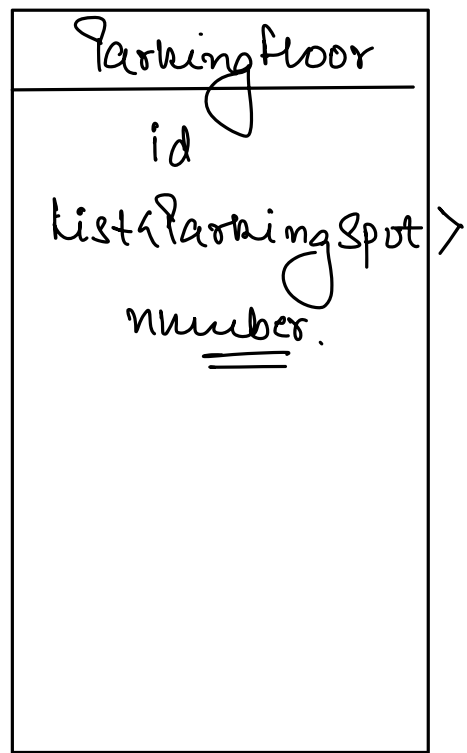
Class Diagram.

Visualization. → User Journey
→ Physical Structure.





→ List<Gate>

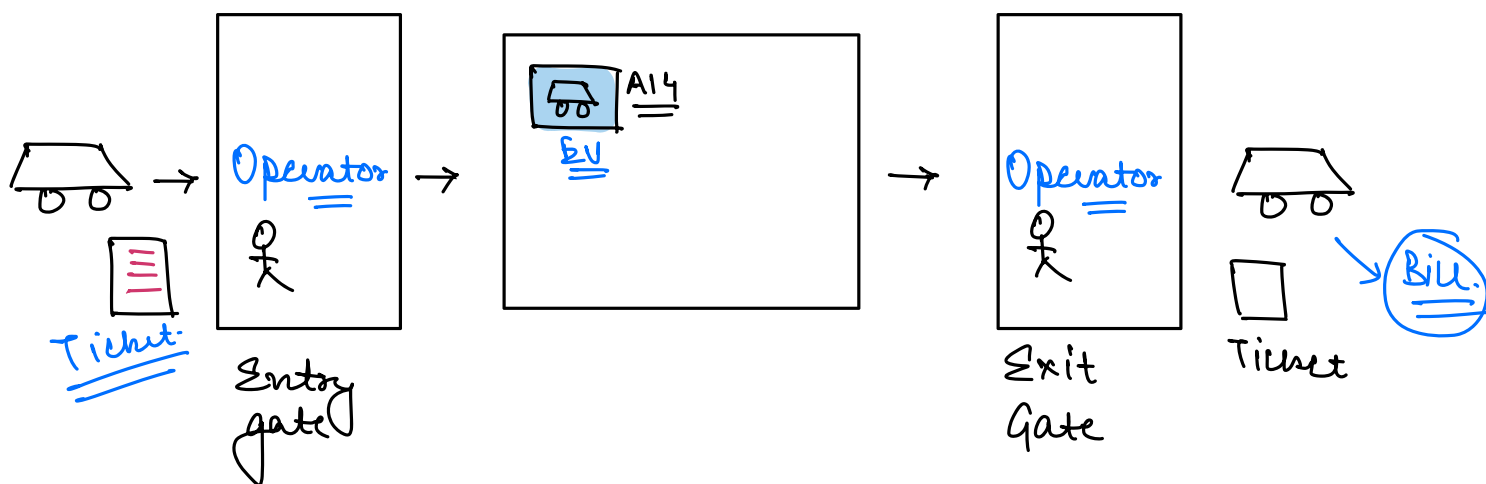


ENTRY

EXIT.

Parking Spot
id
number
Status
VehicleType

Vehicle
id
number
Owner
VehicleType



Operator
id
name
—
—
—

Ticket
id
number
Vehicle
entryTime
Parking Spot
status
Operator
Gate

Bill
id
number
amount
exitTime
ticket
Operator
Gate
Status
list<Payment>

Payment
id
amount
Mode
time
Status
<u>referenceNo.</u>

UPI
Cash
Debit Card
≡

Partial Payment

200 ₹ < 50 ₹ ⇒ Cash
150 ₹ ⇒ UPI

Amazon

↳ 1000 ₹ < 200 ₹ ⇒ Amazon Pay ⇒ P₁
800 ₹ ⇒ Credit Card ⇒ P₂

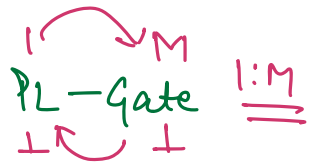
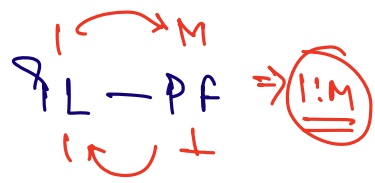
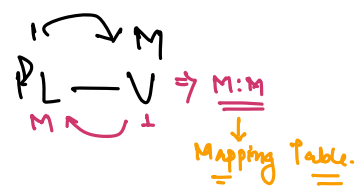
Schema Design.

parking-lots

id	<u>status</u>
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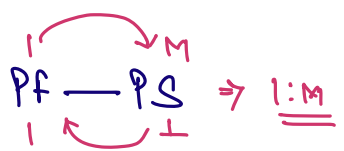
status

id	value
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parking-floors

id	number	parking-lot-id
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parking-spots

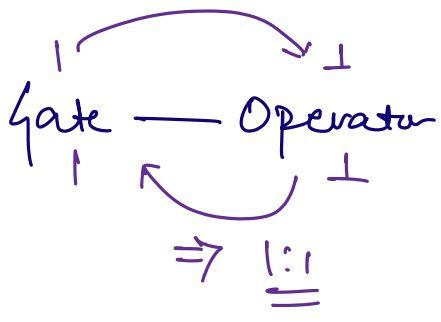
id	number	<u>status</u>	parking-floor-id
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gates

operator-id	parking-lot-id
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gate-status

id	value
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operators

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vehicles

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tickets

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bikes

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payments.

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⇒ Complete Schema Design.

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